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# 1. IMPORT LIBRARIES
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split, RandomizedSearchCV
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import joblib

from google.colab import files
uploaded = files.upload()

# 2. LOAD DATASET
df = pd.read_csv("featured_housing.csv")

print("Data Loaded Successfully")
print("Shape:", df.shape)

# 3. FEATURES & TARGET
X = df.drop("median_house_value", axis=1)
y = df["median_house_value"]

# 4. TRAIN TEST SPLIT
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 5. MODEL
model = RandomForestRegressor(random_state=42)

# 6. PARAMETER DISTRIBUTION (FAST)
param_dist = {
    'n_estimators': [50, 100],
    'max_depth': [10, 20, None],
    'min_samples_split': [2, 5],
    'min_samples_leaf': [1, 2]
}

# 7. RANDOMIZED SEARCH (FAST TUNING)
random_search = RandomizedSearchCV(
    estimator=model,
    param_distributions=param_dist,
    n_iter=10,
    cv=3,
    scoring='r2',
    n_jobs=-1,
    random_state=42,
    verbose=1
)

random_search.fit(X_train, y_train)

# 8. BEST PARAMETERS
print("\nBest Parameters:")
print(random_search.best_params_)

print("\nBest Cross Validation Score (R2):")
print(random_search.best_score_)

# 9. FINAL MODEL (BEST)
best_model = random_search.best_estimator_

# 10. PREDICTION
y_pred = best_model.predict(X_test)

# 11. EVALUATION
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("\nFINAL MODEL PERFORMANCE")

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print("\nFINAL MODEL PERFORMANCE")
print("-----")
print("RMSE:", rmse)
print("MAE :", mae)
print("R2 Score:", r2)
```

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# 12. SAVE MODEL
joblib.dump(best_model, "tuned_house_price_model.pkl")

print("\nTuned model saved successfully!")
```

[Choose files](#) featured_housing.csv

featured_housing.csv(text/csv) - 4339602 bytes, last modified: 17/04/2026 - 100% done

Saving featured_housing.csv to featured_housing.csv

Data Loaded Successfully

Shape: (20640, 22)

Fitting 3 folds for each of 10 candidates, totalling 30 fits

Best Parameters:

{'n_estimators': 100, 'min_samples_split': 2, 'min_samples_leaf': 2, 'max_depth': None}

Best Cross Validation Score (R2):

0.8070173723512525

FINAL MODEL PERFORMANCE

RMSE: 49900.81656042529

MAE : 32132.157622441006

R2 Score: 0.8099760048393496

Tuned model saved successfully!

Start coding or [generate](#) with AI.